Naveen John

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Summary

Software engineer with 3 years across enterprise backend development and applied machine learning. Shipped Java, Python, and Apex integrations into shared enterprise codebases at Cognizant, then built RAG and LLM-evaluation pipelines over electronic health records at UTSA's CARE Lab, and now designs the Python data services and camera–LiDAR ML pipelines behind micro-mobility safety research at ScooterLab. Works end to end: REST APIs and SQL schema design, distributed systems on AWS, Docker, and Kubernetes, and PyTorch/CUDA model training and inference optimization. MS in Computer Science, 3.93 GPA.

Skills

Languages: Python, Java, C++, C, SQL, TypeScript, JavaScript, PHP, Apex, SOQL/SOSL, HTML/CSS
Backend: REST APIs, microservices, system design, ETL pipelines, data modeling, caching, PostgreSQL, MySQL, Redis
Cloud & Infrastructure: AWS (EC2, S3, VPC), Azure, Docker, Kubernetes, Linux, CI/CD, Git, distributed systems, observability
AI/ML: PyTorch, RAG pipelines, LLM evaluation, LangChain, Chroma, Ollama, OpenCV, YOLO, sensor fusion, CUDA, TensorRT
Practices: Agile/Scrum, code review, unit/integration testing, root cause analysis, technical documentation, Jira, ServiceNow
Certifications: Microsoft Azure Certified; 7x Salesforce Certified (Administrator, JavaScript Developer, Platform Developer, AI Associate, Data Cloud)

Experience

Software Engineer, ScooterLab (UTSA), San Antonio, TX — Jul 2025 – Present

- Designed Python ETL pipelines that ingest, time-synchronize, and join 3 data modalities (sensor telemetry, ride video, tabular trip logs) into one curated store, cutting per-study data preparation by 96%
- Built backend REST services over PostgreSQL exposing derived ride features to analytics dashboards, replacing ad-hoc SQL pulls for 7 research teams across 10+ projects and halving average query turnaround
- Instrumented pipelines with schema validation, freshness checks, and alerting across 3 stages, catching data-quality defects before ingest and saving ~6 hrs/week of manual troubleshooting

Graduate Research Assistant, CARE Lab (UTSA), San Antonio, TX — Sep 2024 – Jul 2025

- Built a retrieval-augmented generation (RAG) pipeline in Python over 10,000 electronic health records, serving 2 LLMs (GPT-4.1, DeepSeek v3.1) through Azure AI Foundry to recommend the most frequent hospital procedures from a patient's demographics and medical history
- Designed an LLM evaluation framework on semantic-similarity accuracy metrics, scoring 1,000 responses on semantic similarity to known procedures and surfacing accuracy gaps of up to 9% across gender, race, ethnicity, and age; first-authored the resulting study with Dr. Ke Yang, submitted to AIM 2026

Program Analyst, Cognizant — Mar 2023 – Jul 2024

- Enhanced backend integrations between Salesforce (Apex, SOQL), Java/Python services, and relational databases for an enterprise client, shipping changes into shared codebases owned by 5 teams
- Diagnosed production incidents through logs, SQL queries, and dependency traces, closing 250+ tickets over 16 months with root-cause documentation and regression tests on every fix
- Partnered with product owners to scope and automate recurring workflows, reducing repeat incidents by 90% and removing ~48 hrs/month of manual effort

Projects

Near-Miss Detection via Camera–LiDAR Depth Fusion | *Python, C++, PyTorch, TensorRT, CUDA*

- Fused 2 sensor modalities (monocular camera, LiDAR point clouds) to refine depth estimation, training models against LiDAR ground truth to correct systematic monocular bias and cut depth RMSE by 80% in urban scenes
- Prototyped the end-to-end inference path in C++ and Python for edge deployment on scooter onboard compute; validated on 3 metrics (RMSE, MAE, time-to-collision accuracy)

UTSA FindMySpot | *Python, OpenCV, YOLO11 — 2nd Place Overall, RowdyHacks XI*

- Built a real-time occupancy pipeline in Python and OpenCV that classifies parking bays per existing CCTV feed at 10 FPS using ROI extraction, background modeling, and per-bay binary classification
- Fine-tuned a pretrained YOLO11 on 1,000+ annotated lot images spanning day/night and rain/clear conditions, reaching 90% detection accuracy after denoising and contrast normalization cut false positives by 90%

RAMP – Research Data Export Platform | *Laravel (PHP), MySQL, REST APIs*

- Shipped 5 backend REST endpoints and the matching front-end views for managing and exporting research datasets, with input validation, structured error handling, and request logging that cut support requests by 10%
- Worked in a 2-person team through Git, code review, and issue tracking, merging 100+ pull requests with zero rollbacks

Education

University of Texas at San Antonio — Aug 2024 – May 2026
Master of Science, Computer Science — *GPA: 3.93 / 4.00*
Coursework: Machine Learning, Data Mining, Data Science, Cloud Computing, Parallel Processing, Analysis of Algorithms, Operating Systems, Computer Architecture